GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Computational Mathematics

DS-502-MJ

Interview Preparation — Topic-wise Q&A

Semester I · Core

Prepared by Siddheshwar Thaware · Free for SPPU students

Curated interview questions and answers for Computational Mathematics, organised by topic from the SPPU M.Sc. Data Science syllabus. Ideal for viva, placement and internship preparation.

Tip: read the answer, then close the PDF and try to explain each concept aloud in your own words.

Linear algebra

Q. What is a vector space?

A set of vectors closed under addition and scalar multiplication, satisfying axioms like associativity, existence of a zero vector and inverses.

Q. What are eigenvalues and eigenvectors?

For matrix A, a non-zero vector v is an eigenvector if Av = ;·`, where ;² —2 F†R V–vVàvalue. They describe directions that are only scaled by the transformation.

Q. What is the rank of a matrix?

The number of linearly independent rows/columns; it equals the dimension of the column space.

Q. What is Singular Value Decomposition (SVD)?

Factorisation A = U:5aÔ v†W&R P, V are orthogonal and :2 †öÆG2 6–æwVÆ " `alues. Used for dimensionality reduction, compression and Latent Semantic Indexing.

Q. Why is linear algebra important in data science?

Data is represented as matrices; operations like PCA, SVD, and neural-network computations are all linear-algebra based.